

Ladder Fuel Risk Mapping from Airborne LiDAR Using Vertical Return-Density Signals and Threshold Detection

Ian Toyota
ES 156 Final Project

Abstract—Ladder fuels are vertically continuous vegetation structures that can carry fire from the surface into the canopy. This project studies ladder-fuel mapping as a one-dimensional signal detection problem derived from airborne light detection and ranging (LiDAR) point clouds. Each horizontal forest cell is represented by a discrete vertical return-density signal, and ladder-fuel risk is estimated from energy in the 2–4 m height band. Three detectors are implemented: a band-ratio statistic, a smoothed band-ratio statistic, and a matched-filter statistic learned from synthetic point clouds. Synthetic experiments with known ground truth yield receiver operating characteristic (ROC) curves with area under the curve near 0.85, demonstrating useful but imperfect separability under realistic class overlap. The same detector pipeline is then applied to a NEON airborne LiDAR tile from the GUAN site, producing a spatial ladder-fuel risk map over 893 valid 30 m cells. While the result is not a full wildfire-fuel model, it is a reproducible signal-processing framework for turning LiDAR returns into interpretable ladder-fuel evidence.

I. INTRODUCTION

A. Motivation

Wildfire is not a single kind of disaster. Low-intensity surface fires can play a beneficial ecological role by consuming litter, brush, and dead wood. Crown fires are different. In a crown fire, combustion moves into the living canopy, often spreading faster and causing more severe and lasting damage than a surface fire under comparable conditions [4]. The transition from surface fire to crown fire is strongly controlled by vertical fuel continuity: shrubs, saplings, and low branches can act as ladder fuels that carry flames from the forest floor into the overstory [5]. This makes the geometry of vegetation not merely descriptive but a fundamental part of the fire’s mechanism.

The human stakes are severe. The U.S. Fire Administration’s analysis of six major wildfires from 2016–2018 reports 85 deaths and public-source losses of about \$8.47 billion for the Camp Fire alone, and about \$2.93 billion in losses for the Woolsey Fire [6]. NOAA’s billion-dollar disaster database reports that the 2020 U.S. wildfire disaster caused \$19.9 billion in damage and 46 deaths [7]. While this already sounds bad, direct fire fatalities are only part of the pain. A 2024 *Science Advances* study estimated that wildfire-attributable PM_{2.5} exposure in California from 2008–2018 caused 52,480–55,710 premature deaths, corresponding to an economic impact of \$432–\$456 billion [8].

Just like human life, the ecological stakes are very high. The 2019–2020 Australian megafires burned 10.3 million hectares and triggered major biodiversity monitoring across more than 2,000 taxa [9]. That study found that wildfire impacts were worse in areas affected by high severity, extensive burning, drought, and recent past fires, with mammals showing especially large responses. Other work on the same megafires found broad habitat impacts for Australian fauna, illustrating that extreme fires can become biodiversity crises rather than isolated disturbance events [10]. This is the sense in which ladder-fuel mapping is engineering for good: better maps do not extinguish fires by themselves, but they can help managers identify where fuel treatments may reduce the chance that an ordinary surface fire becomes a catastrophic canopy fire.

B. Overview

Airborne LiDAR is a pulsed electromagnetic remote-sensing system. A laser pulse is transmitted, scattered by terrain and vegetation, and received after a propagation delay. If a return arrives at time delay τ , the corresponding range is approximately

$$r = \frac{c\tau}{2}, \quad (1)$$

where c is the speed of light. After georeferencing many returns, the measurement becomes a three-dimensional point cloud. For forest applications, the point cloud is often more useful after converting absolute elevation to height above ground.

This project asks a focused question: can ladder-fuel risk be estimated from the vertical distribution of LiDAR returns? Rather than attempting a complete operational wildfire-fuel classifier, the project treats each grid cell as a discrete-time signal in height. The “time” variable is replaced by vertical height, and the signal value is return density. The problem then resembles the signal-detection problems studied in signals and systems: construct a statistic from sampled data, compare it to a threshold, and quantify the miss/false-alarm tradeoff.

The project has three parts. First, a signal model is developed for vertical LiDAR return density. Second, synthetic point clouds are generated with known low/high ladder-fuel labels, allowing controlled ROC analysis. Third, the detector is applied to real airborne LiDAR data from the National Ecological Observatory Network (NEON).

C. Related Work

The signal-processing language follows Oppenheim, Will-sky, and Nawab's treatment of signals, systems, sampling, convolution, and frequency-domain reasoning [1]; the numerical presentation is also informed by the application-oriented style of Chaparro and Akan [2]. The smoothing operation used here is a short finite impulse response (FIR) filter, and the matched-filter detector is a direct application of correlation-based detection. For statistical detection and ROC evaluation, this work follows the standard hypothesis-testing viewpoint in Kay [3].

LiDAR has long been used for terrain and vegetation remote sensing because it directly samples three-dimensional structure [11]. This matters because traditional fuel descriptions are limited exactly where ladder fuels are most important. Field measurements are local and labor intensive, while common canopy-base-height surrogates can miss shrubs and small trees that still help carry fire upward [12]. Kramer *et al.* showed that LiDAR-derived percent cover in the 2–4 m band was useful for distinguishing ladder-fuel conditions [12]; a follow-up study combined field photography and LiDAR and emphasized that existing ladder-fuel quantification methods remain limited and imprecise [13]. These observations motivate the principal statistic in this project. The real-data demonstration uses NEON's discrete-return LiDAR point-cloud product, which provides classified airborne LiDAR returns in standard LAS/LAZ format [14].

D. Notation

Let i index a horizontal grid cell, and let z denote height above ground. The continuous vertical return density in cell i is denoted $x_i(z)$. After binning heights with bin width Δz , the sampled signal is

$$x_i[n] = \#\{\text{returns in cell } i \text{ with } z \in [n\Delta z, (n+1)\Delta z)\}. \quad (2)$$

The bin center is denoted z_n . The vegetation-normalized signal is

$$\tilde{x}_i[n] = \frac{x_i[n]}{\sum_{m: z_m \geq z_{\min}} x_i[m]}, \quad (3)$$

where $z_{\min} = 0.5$ m removes ground returns from the normalization. The binary labels are H_0 for low ladder-fuel loading and H_1 for high ladder-fuel loading. A detector score is denoted $T(x_i)$ and is compared to threshold γ .

II. PART I: LiDAR SIGNAL MODEL

A LiDAR waveform can be idealized as a transmitted pulse $s(t)$ reflected by a vertical distribution of scatterers. If a cell contains scattering surfaces at delays τ_k with amplitudes a_k , then the received signal may be written as

$$y(t) = \sum_k a_k s(t - \tau_k) + w(t), \quad (4)$$

where $w(t)$ models receiver noise, background light, and unmodeled scattering. In a discrete-return point cloud, peak extraction has already converted the waveform into range

samples. This project begins after that extraction step and studies the sampled vertical structure.

For each 30 m cell, returns are binned into a vertical signal $x_i[n]$ with $\Delta z = 0.25$ m. The ladder-fuel band is defined as

$$\mathcal{B} = \{n : 2 \leq z_n < 4\}. \quad (5)$$

The simplest detector is the band-ratio statistic

$$T_B(x_i) = \frac{\sum_{n \in \mathcal{B}} x_i[n]}{\sum_{n: z_n \geq 0.5} x_i[n]}. \quad (6)$$

The 2–4 m interval is a modeling choice rather than a universal ecological constant. It is selected because prior ladder-fuel studies found LiDAR-derived cover in this band to be informative for distinguishing ladder-fuel conditions [12], and because vegetation in this height range is high enough to bridge surface fuels toward the lower canopy while still being visible in airborne LiDAR returns. In other forest types or management settings, the same framework could be applied with a shifted or widened band, or with several height bands used jointly.

This is a vertical-energy detector: it measures how much of the non-ground return mass lies in the height interval associated with ladder fuels.

The second detector smooths the vertical signal before applying (6). The FIR filter is

$$h[n] = \frac{1}{9}[1, 2, 3, 2, 1], \quad (7)$$

and the smoothed signal is $x_i^{(s)}[n] = (h * x_i)[n]$. Smoothing reduces sensitivity to bin-boundary effects and small height perturbations.

The third detector is a matched-filter statistic. From synthetic training data, class-mean normalized signals $\mu_0[n]$ and $\mu_1[n]$ are estimated. The template is

$$q[n] = \frac{\mu_1[n] - \mu_0[n]}{\|\mu_1 - \mu_0\|_2}, \quad (8)$$

with entries below 0.5 m set to zero. The matched-filter score is

$$T_M(x_i) = \sum_n \tilde{x}_i[n] q[n]. \quad (9)$$

The resulting detector has the familiar correlation form: cells are scored highly when their vertical return-density shape resembles the difference between high- and low-fuel training profiles.

III. PART II: SYNTHETIC DETECTION EXPERIMENT

A. Synthetic Point-Cloud Model

The synthetic model generates independent grid cells under H_0 and H_1 . Each cell contains a random number of returns sampled from a mixture of components: ground, low shrub, ladder-fuel vegetation, lower canopy, upper canopy, and sparse outlier returns. The two classes differ mainly in the probability mass assigned to the 2–4 m ladder-fuel component. Random mixture weights, height noise, and point-density variation are included so that the classes overlap.

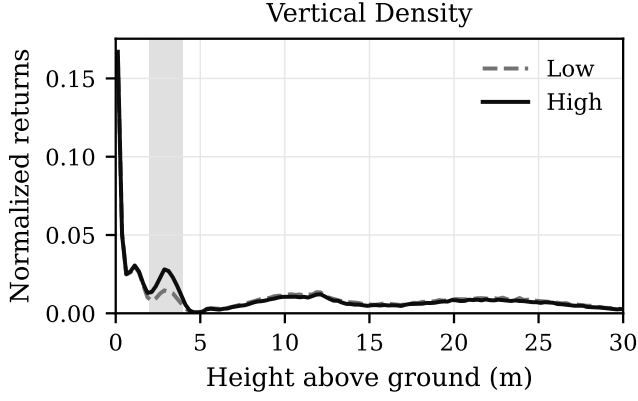


Fig. 1. Class-mean synthetic vertical return-density signals. The shaded band indicates the 2–4 m interval used for ladder-fuel detection.

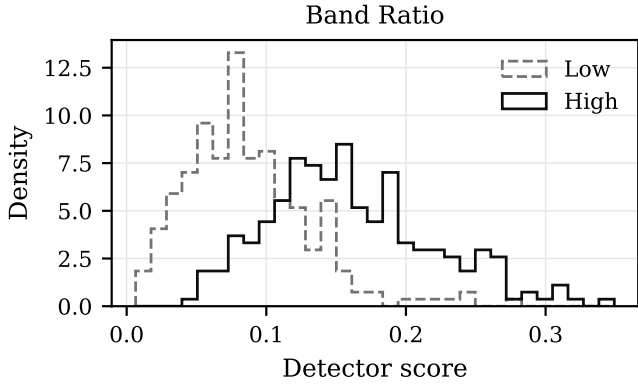


Fig. 2. Band-ratio detector scores for low and high synthetic ladder-fuel classes. The overlap between the distributions creates the false-alarm/miss tradeoff.

This synthetic stage is essential because real public LiDAR tiles rarely provide cell-level ladder-fuel labels. Synthetic data supplies controlled ground truth, enabling quantitative estimates of detection probability and false-alarm probability before the method is transferred to a real tile.

B. Threshold Detection and ROC Curves

Each detector computes a scalar score $T(x_i)$. A cell is declared high ladder fuel if

$$T(x_i) \geq \gamma. \quad (10)$$

For a threshold γ , the detection and false-alarm probabilities are

$$P_D(\gamma) = \Pr(T(X) \geq \gamma \mid H_1), \quad (11)$$

$$P_{FA}(\gamma) = \Pr(T(X) \geq \gamma \mid H_0). \quad (12)$$

The ROC curve plots P_D versus P_{FA} as γ varies. The operating threshold reported in Table I is selected on the training set by maximizing $P_D - P_{FA}$ and then evaluated on a held-out test set.

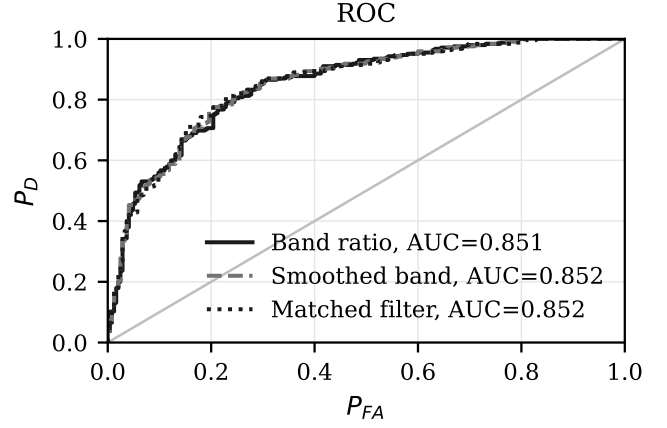


Fig. 3. ROC curves for the three synthetic detectors. AUC values near 0.85 indicate useful but imperfect separability.

TABLE I
SYNTHETIC TEST-SET DETECTION RESULTS

Detector	AUC	P_D	P_{FA}	Acc.
Band ratio	0.851	0.669	0.151	0.759
Smoothed band	0.852	0.780	0.229	0.776
Matched filter	0.852	0.780	0.224	0.778

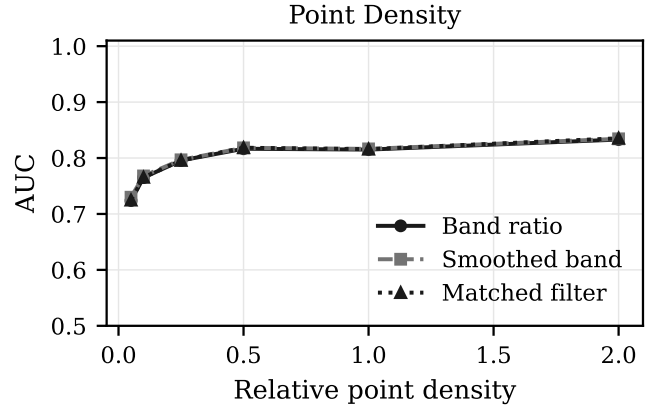


Fig. 4. AUC under different relative point densities. The performance drop at sparse densities shows the sampling dependence of the vertical-density signal.

The results show a deliberate middle ground. The detector is clearly better than random guessing, but the problem is not artificially separable. This is important: if the synthetic data produced AUC equal to one, the simulation would say more about the convenience of the generator than the difficulty of ladder-fuel detection. The current model forces the detector to handle class overlap, random canopy structure, and finite sampling.

Fig. 4 studies sampling density. At very low relative point density, the vertical histogram is a noisy estimate of the underlying return-density signal. As point density increases, AUC improves and then levels off, suggesting that class

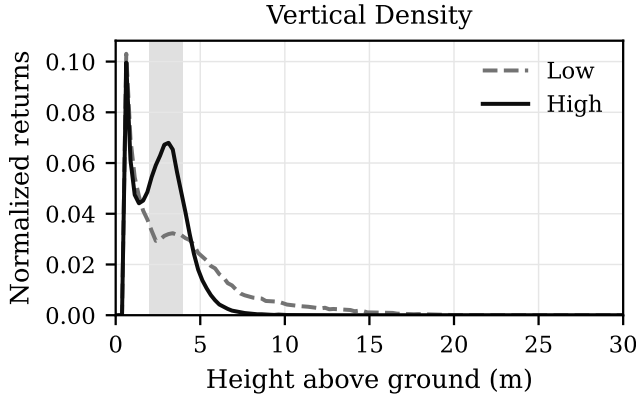


Fig. 5. NEON vertical return-density profiles after splitting valid cells at the median smoothed-band score. This plot is descriptive, not a ground-truth validation.

variability rather than sampling noise becomes the limiting factor.

IV. PART III: NEON REAL-DATA APPLICATION

A. Data and Preprocessing

The real-data demonstration uses NEON product DP1.30003.001, the discrete-return airborne LiDAR point cloud. The selected file is a classified LAZ tile from the GUAN site in May 2018:

```
NEON_D04_GUAN_DP1_724000_1986000
_classified_point_cloud_colorized.laz
```

The tile contains approximately 5.2×10^6 returns. For a fast and reproducible project pipeline, the results reported here use a deterministic random subsample of up to 1.2×10^6 returns before height normalization and gridding.

The classified ground returns are used to estimate a ground surface by nearest-neighbor interpolation. Height above ground is then computed for each return. Returns below 0.5 m are excluded from detector scoring so that ground points do not dominate the vertical-density signal. The tile is divided into 30 m cells, and cells with fewer than 80 non-ground returns are discarded. This yields 893 valid cells.

B. Risk Map

The same detector used in the synthetic experiment is applied to each valid NEON cell. Fig. 6 shows the smoothed-band score over the tile. Darker cells have larger fractions of non-ground LiDAR returns in the 2–4 m ladder-fuel band. The mean smoothed-band score over valid cells is 0.379, and the 90th percentile is 0.550.

Because no field-validated ladder-fuel labels are used for this tile, Fig. 6 should not be interpreted as a verified fire-behavior map. It is instead a real-data transfer test where a detector trained on synthetic ground truth is applied to airborne LiDAR to generate a spatially interpretable score. The result highlights structural vegetation within a ladder-fuel height

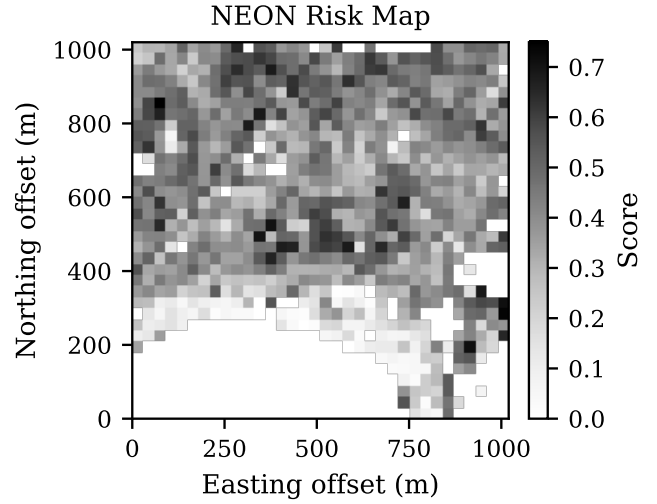


Fig. 6. Ladder-fuel risk map for the NEON GUAN tile. The score is the smoothed fraction of non-ground returns in the 2–4 m band.

band, but it does not capture other factors such as ignition potential, moisture, wind, or suppression difficulty.

The detector also treats grid cells independently. This is appropriate for the scope of the present work because the goal is to estimate local vertical fuel structure from the LiDAR signal before modeling fire spread. In reality, vegetation is spatially correlated and fire propagates across neighboring cells. A more robust mapping stage could therefore smooth detector scores over adjacent cells, for example

$$\bar{T}_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} T_j, \quad (13)$$

where $\mathcal{N}(i)$ is a neighborhood around cell i . This would reduce isolated noisy detections and better reflect spatial continuity, but it would also mix the signal-processing detector with a spatial fire-behavior model. Keeping the cells independent makes the present result easier to interpret: each score is evidence from that cell's vertical LiDAR return-density signal alone.

V. CONCLUSION

This project demonstrates that ladder-fuel detection can be effectively framed as a signal processing problem. By representing LiDAR point cloud data as vertical return-density signals, we constructed and evaluated a set of detectors ranging from simple band-energy ratios to a learned matched filter. Results on synthetic data show that increasing model structure improves detection performance, and application to real airborne LiDAR produces spatially interpretable outputs.

Importantly, this approach enables a clean separation between structural signal detection and broader ecological interpretation. The detectors developed here identify statistical evidence of vegetation within a specified height band, rather than providing a complete assessment of wildfire risk. Factors

such as fuel moisture, wind conditions, and ignition likelihood remain outside the scope of this model.

Future work could extend this framework in several directions. One practical direction is the development of a lightweight pipeline or wrapper that allows users to input LiDAR data and directly obtain ladder-fuel classifications, making the method more accessible for applied use. Additional extensions include incorporating spatial context between neighboring cells, adapting the detector to different vegetation regimes, and validating performance against field data. Together, these steps would help bridge the gap between controlled detection results and real-world fire management applications.

CODE AVAILABILITY

All simulations, detector implementations, and figure-generation scripts are available in the public project repository.

The primary entry points for reproducing results are:

- `scripts/run_synthetic_experiments.py`
- `scripts/download_neon_tile.py`
- `scripts/run_neon_demo.py`

REFERENCES

- [1] A. V. Oppenheim, A. S. Willsky, and S. H. Nawab, *Signals and Systems*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 1997.
- [2] L. F. Chaparro and A. Akan, *Signals and Systems Using MATLAB*, 3rd ed. London, U.K.: Academic Press, 2019.
- [3] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory*. Upper Saddle River, NJ, USA: Prentice Hall, 1998.
- [4] J. H. Scott and E. D. Reinhardt, "Estimating canopy fuels in conifer forests," *Fire Management Today*, vol. 62, no. 4, pp. 45–50, 2002.
- [5] National Park Service, "Wildland fire spread and suppression," 2025. [Online]. Available: <https://www.nps.gov/articles/wildland-fire-spread-and-suppression.htm>
- [6] U.S. Fire Administration, *An Analysis of NFIRS Data for Selected Wildfires*. Emmitsburg, MD, USA: FEMA, 2022.
- [7] NOAA National Centers for Environmental Information, "Billion-dollar weather and climate disasters: 2020 summary stats," 2025. [Online]. Available: <https://www.ncei.noaa.gov/access/billions/summary-stats/US-2020.html>
- [8] R. Connolly *et al.*, "Mortality attributable to PM_{2.5} from wildland fires in California from 2008 to 2018," *Science Advances*, vol. 10, no. 23, Art. no. ead11252, 2024.
- [9] D. A. Driscoll *et al.*, "Biodiversity impacts of the 2019–2020 Australian megafires," *Nature*, vol. 635, pp. 898–905, 2024.
- [10] M. Ward *et al.*, "Impact of 2019–2020 mega-fires on Australian fauna habitat," *Nature Ecology & Evolution*, vol. 4, pp. 1321–1326, 2020.
- [11] E. P. Baltsavias, "Airborne laser scanning: basic relations and formulas," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 54, no. 2–3, pp. 199–214, 1999.
- [12] H. A. Kramer, B. M. Collins, M. L. Kelly, and S. L. Stephens, "Quantifying ladder fuels: a new approach using LiDAR," *Forests*, vol. 5, no. 6, pp. 1432–1453, 2014.
- [13] H. A. Kramer, B. M. Collins, F. K. Lake, M. K. Jakubowski, S. L. Stephens, and M. Kelly, "Estimating ladder fuels: a new approach combining field photography with LiDAR," *Remote Sensing*, vol. 8, no. 9, Art. no. 766, 2016.
- [14] National Ecological Observatory Network, "Discrete return LiDAR point cloud, DP1.30003.001," NEON Data Portal. [Online]. Available: <https://data.neonscience.org/data-products/DP1.30003.001>